



REVA
UNIVERSITY
Bengaluru, India

Department of Computer Science and Engineering

Mini Project - B22CI0605

Title of the Project : **NeuroVidya**: An AI-Powered Adaptive Learning Platform
for Dyslexic Students

Presented By : Group 42

HARISH CHAWAN A - R23EK010

JAGADEEP H M - R23EK012

KEERTHI K - R23EK014

ABEL BIJU PAREL - R23EK056

Guided By : Dr Harsha B K

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1. Introduction

What is NeuroVidya?

- AI-powered adaptive learning platform for students with dyslexia
- Name combines 'Neuro' (neurodiversity) + 'Vidya' (knowledge)

Vision:

'A calm digital learning space for neurodiverse minds'

Key Features:

- Adaptive Reading - Customizable fonts, spacing, focus mode
- AI Assistance - Text simplification, explanations
- Visual Games - Interactive learning through play
- Progress Tracking - Analytics, streaks, achievements



2. Literature Review

Understanding Dyslexia:

- Affects 10-20% of global population
- 1 in 5 students has language-based learning disability
- Often undiagnosed until late primary school

Challenges Faced:

- Letter recognition > Reading delays
- Phonological processing > Spelling difficulties
- Visual processing > Word confusion (b/d, p/q)
- Working memory > Comprehension challenges

Gaps in Existing Solutions:

- Lack of integrated platforms
- Minimal AI-powered personalization
- Limited visual learning components

Research Foundation:

- International Dyslexia Association guidelines
- Multi-sensory learning theory (VARK model)
- Universal Design for Learning (UDL) principles

3. Problem Statement

Core Problem:

Students with dyslexia struggle with traditional education platforms not designed for their unique learning needs.

Specific Problems:

1. Reading Barriers

- Standard fonts difficult to process
- Reading fatigue, comprehension issues, avoidance behavior

2. Limited Writing Support

- 'becuse' not caught by standard spell check

3. No Visual Learning

- Most platforms are text-heavy

4. Progress Tracking Gaps

- No personalized learning paths

The Need: A comprehensive, accessible platform with multi-sensory learning that adapts to individual abilities.

4. Objectives

1. Dyslexia-Friendly Reading Interface

- Customizable fonts (Lexend, OpenDyslexic)
- Adjustable spacing and focus mode
- Text-to-speech with word highlighting

2. AI-Powered Learning Assistance

- Text simplification by reading level
- Intelligent spell-checking for dyslexic patterns
- AI chatbot for concept explanations

3. Visual Learning Games

- Word-Image Matching, Letter Recognition
- Syllable Builder, Sentence Builder
- Illustrated Stories, Visual Concepts

4. Progress Tracking

- Reading speed, spelling accuracy
- Learning streaks, visual analytics

5. Software Requirements

FRONTEND:

- React 18.2+ - UI Framework
- TypeScript 5.3+ - Type Safety
- Vite 5.0+ - Build Tool
- Tailwind CSS 3.3+ - Styling
- Tesseract.js 5.0+ - OCR Processing
- React Router 6.20+ - Navigation

BACKEND:

- Python 3.11+ - Runtime
- FastAPI (Latest) - API Framework
- PostgreSQL 14+ - Database

EXTERNAL SERVICES:

- OpenAI API / Anthropic Claude - AI processing
- Web Speech API - Text-to-speech & Speech Recognition

6. Methodology

Development Process:

RESEARCH > DESIGN > DEVELOPMENT > TESTING > ITERATION

Technical Architecture:

Frontend (React + TypeScript)

↓ HTTP/REST

Backend (FastAPI + Python)

↓

Database (PostgreSQL)

Database Schema:

- Users - Authentication and profiles
- ReadingPreferences - Customization settings
- LearningProgress - Metrics and streaks
- GameScores - Game performance data
- ActivityLog - User activity tracking

Key Algorithms:

1. Dyslexia Spell Correction - Pattern matching + Edit distance
2. Focus Mode - Sentence highlighting + Text dimming
3. Streak Calculation - 48-hour grace period

7. Implementation

Phase 1: Foundation [DONE]

- Project setup, Backend, Database schema, Design system, Auth

Phase 2: Core Features [DONE]

- DyslexiaText, ReadingDisplay, TTSControl, OCRScanner, FocusMode

Phase 3: AI Integration [DONE]

- Text simplification, Concept explanation, Spell checking, AI chat

Phase 4: Visual Games [DONE]

- Word-Image Matching, Letter Recognition, Syllable Builder
- Sentence Builder, Illustrated Stories, Visual Concepts

Phase 5: Analytics [DONE]

- StatsOverview, ReadingChart, GameScoresChart, StreakDisplay

API Endpoints Implemented:

/api/auth/*, /api/text/*, /api/assistant/*, /api/games/*, /api/analytics/*

8. Applications

Primary Applications:

Educational Institutions:

- Special education classrooms, Resource rooms
- After-school tutoring, Library accessibility services

Individual Learners:

- Self-directed learning, Homework assistance
- Exam preparation support

Therapeutic Settings:

- Speech therapy, Occupational therapy
- Educational therapy programs

Use Case Examples:

1. Daily Reading - Upload > Format > Read with TTS > Track progress
2. Homework Help - Struggle > Explain > Simplified > Complete
3. Visual Learning - Play games > Improve recognition > Apply skills

Target Segments: Elementary (6-10), Middle (11-14)

9. Expected Conclusion

Anticipated Outcomes:

For Learners:

- Reading fluency: 20-30% improvement expected
- Spelling accuracy: 15-25% improvement
- Increased confidence and independence
- More voluntary reading

For Educators:

- Data-driven insights into student progress
- Tools for personalized instruction

Success Metrics:

- Reading speed: 20-30% increase in 3 months
- User retention: 70%+ over academic year
- Feature satisfaction: 4.5/5+ rating

Conclusion:

'NeuroVidya represents a new approach to inclusive education leveraging modern technology to create learning experiences that adapt to the learner, not the other way around.'

10. References

Academic References:

1. International Dyslexia Association. (2023). Dyslexia in the Classroom
2. Shaywitz, S. (2020). Overcoming Dyslexia (2nd ed.). Knopf
3. Rose, J. (2009). Identifying and Teaching Children with Dyslexia

Technical Documentation:

- React: <https://react.dev>
- FastAPI: <https://fastapi.tiangolo.com>
- OpenAI API: <https://platform.openai.com>

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Thank You